

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

Contents

1	Tuesday 16 June 2026	2
1.1	Daily Limit	2
2	Wednesday 17 June 2026	3
2.1	Daily Limit	3
3	Thursday 18 June 2026	4
3.1	Daily Limit	4
4	Friday 19 June 2026	5
4.1	Daily Limit	5
5	Saturday 20 June 2026	7
5.1	Daily Limit	7

Tuesday 16 June 2026

1.1 Daily Limit

Limit of an Exponential Trigonometric Function

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{e^{\sin(x)} - e^{\tan(x)}}{x^3}$$

Solution: To evaluate the limit, we begin by factoring out $e^{\sin(x)}$ from the numerator:

$$\lim_{x \rightarrow 0} \frac{e^{\sin(x)} \left(1 - e^{\tan(x) - \sin(x)}\right)}{x^3}$$

Since the limit of a product is the product of the limits (provided they exist), we can evaluate the limit of $e^{\sin(x)}$ as $x \rightarrow 0$ separately. As $\sin(0) = 0$, $e^{\sin(0)} = e^0 = 1$. The limit thus simplifies to:

$$\lim_{x \rightarrow 0} \frac{1 - e^{\tan(x) - \sin(x)}}{x^3}$$

Recall the standard limit $\lim_{u \rightarrow 0} \frac{e^u - 1}{u} = 1$. By letting $u = \tan(x) - \sin(x)$, we note that $u \rightarrow 0$ as $x \rightarrow 0$. We can rewrite the expression by multiplying and dividing by u :

$$\lim_{x \rightarrow 0} \left[\frac{1 - e^{\tan(x) - \sin(x)}}{\tan(x) - \sin(x)} \cdot \frac{\tan(x) - \sin(x)}{x^3} \right]$$

The first factor approaches -1 . For the second factor, we express $\tan(x)$ in terms of sine and cosine:

$$\frac{\tan(x) - \sin(x)}{x^3} = \frac{\frac{\sin(x)}{\cos(x)} - \sin(x)}{x^3} = \frac{\sin(x) \left(\frac{1}{\cos(x)} - 1 \right)}{x^3} = \frac{\sin(x)(1 - \cos(x))}{x^3 \cos(x)}$$

Now, we can split this into familiar standard limits:

$$\lim_{x \rightarrow 0} \left[\frac{\sin(x)}{x} \cdot \frac{1 - \cos(x)}{x^2} \cdot \frac{1}{\cos(x)} \right]$$

Evaluating each part as $x \rightarrow 0$:

- $\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$
- $\lim_{x \rightarrow 0} \frac{1 - \cos(x)}{x^2} = \frac{1}{2}$
- $\lim_{x \rightarrow 0} \frac{1}{\cos(x)} = 1$

Thus, the limit of the second factor evaluates to $1 \cdot \frac{1}{2} \cdot 1 = \frac{1}{2}$. Multiplying this by the -1 from the first factor gives the final result:

$$\lim_{x \rightarrow 0} \frac{e^{\sin(x)} - e^{\tan(x)}}{x^3} = -1 \cdot \frac{1}{2} = -\frac{1}{2}$$

Wednesday 17 June 2026

2.1 Daily Limit

Limit Involving Exponentials and Infinity

Evaluate the limit:

$$\lim_{x \rightarrow \infty} x^2 \left(e^{\frac{1}{x+1}} - e^{\frac{1}{x}} \right)$$

Solution: To simplify the behavior at infinity, let us make the substitution $t = \frac{1}{x}$. As $x \rightarrow \infty$, $t \rightarrow 0^+$. The term $\frac{1}{x+1}$ can be rewritten in terms of t :

$$x + 1 = \frac{1}{t} + 1 = \frac{1+t}{t} \implies \frac{1}{x+1} = \frac{t}{1+t}$$

Substituting these into the limit expression gives:

$$\lim_{t \rightarrow 0^+} \frac{e^{\frac{t}{1+t}} - e^t}{t^2}$$

Factor out e^t from the numerator:

$$\lim_{t \rightarrow 0^+} e^t \left(\frac{e^{\frac{t}{1+t} - t} - 1}{t^2} \right)$$

Since $e^t \rightarrow 1$ as $t \rightarrow 0^+$, we can drop it from the evaluation and focus on the rest. We first simplify the exponent:

$$\frac{t}{1+t} - t = \frac{t - t(1+t)}{1+t} = \frac{-t^2}{1+t}$$

The limit becomes:

$$\lim_{t \rightarrow 0^+} \frac{e^{-\frac{t^2}{1+t}} - 1}{t^2}$$

We multiply and divide by the exponent $-\frac{t^2}{1+t}$ to utilize the standard limit form $\frac{e^u - 1}{u} \rightarrow 1$ as $u \rightarrow 0$:

$$\lim_{t \rightarrow 0^+} \left[\frac{e^{-\frac{t^2}{1+t}} - 1}{-\frac{t^2}{1+t}} \cdot \frac{-\frac{t^2}{1+t}}{t^2} \right]$$

As $t \rightarrow 0^+$, the exponent $-\frac{t^2}{1+t} \rightarrow 0$, so the first factor evaluates to 1. The remaining fraction simplifies algebraically:

$$\lim_{t \rightarrow 0^+} \frac{-t^2}{t^2(1+t)} = \lim_{t \rightarrow 0^+} \frac{-1}{1+t} = \frac{-1}{1+0} = -1$$

Therefore, the total limit evaluates to:

$$-1$$

Thursday 18 June 2026

3.1 Daily Limit

Limit of Trigonometric and Hyperbolic Functions

Given the function $f(x) = \frac{\sin(\sqrt{x}) \sin^{-1}(x^2)}{\sinh(\sqrt{x}) \sinh^{-1}(x^2)}$, find:

$$\lim_{x \rightarrow 0} f(x)$$

Solution: This limit can be evaluated by leveraging the standard fundamental limits of trigonometric and hyperbolic functions and their inverses near zero. We recall the following properties as $u \rightarrow 0$:

$$\lim_{u \rightarrow 0} \frac{\sin(u)}{u} = 1, \quad \lim_{u \rightarrow 0} \frac{\sin^{-1}(u)}{u} = 1, \quad \lim_{u \rightarrow 0} \frac{\sinh(u)}{u} = 1, \quad \lim_{u \rightarrow 0} \frac{\sinh^{-1}(u)}{u} = 1$$

We can rewrite the given function $f(x)$ by dividing and multiplying by the respective arguments of each function. For $f(x)$, the arguments are $\sqrt{x}$ and x^2 . Both of these arguments approach 0 as $x \rightarrow 0$.

Let's restructure the expression:

$$f(x) = \left(\frac{\sin(\sqrt{x})}{\sqrt{x}} \right) \cdot \left(\frac{\sqrt{x}}{\sinh(\sqrt{x})} \right) \cdot \left(\frac{\sin^{-1}(x^2)}{x^2} \right) \cdot \left(\frac{x^2}{\sinh^{-1}(x^2)} \right)$$

Notice that the inserted factors $\sqrt{x}$ and x^2 perfectly cancel out across the numerator and denominator, so the expression remains mathematically equivalent to the original $f(x)$.

Now, we take the limit as $x \rightarrow 0$. Applying the limit to the product of these functions allows us to evaluate each term independently:

$$\lim_{x \rightarrow 0} \frac{\sin(\sqrt{x})}{\sqrt{x}} = 1$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{x}}{\sinh(\sqrt{x})} = \left(\lim_{x \rightarrow 0} \frac{\sinh(\sqrt{x})}{\sqrt{x}} \right)^{-1} = 1^{-1} = 1$$

$$\lim_{x \rightarrow 0} \frac{\sin^{-1}(x^2)}{x^2} = 1$$

$$\lim_{x \rightarrow 0} \frac{x^2}{\sinh^{-1}(x^2)} = \left(\lim_{x \rightarrow 0} \frac{\sinh^{-1}(x^2)}{x^2} \right)^{-1} = 1^{-1} = 1$$

Multiplying these individual limits together gives:

$$\lim_{x \rightarrow 0} f(x) = 1 \cdot 1 \cdot 1 \cdot 1 = 1$$

Friday 19 June 2026

4.1 Daily Limit

Limit Involving Factorials and Exponents

Find the integer nearest to the value of the limit:

$$\lim_{n \rightarrow \infty} \left(\frac{(2n)!}{n!n^n} \right)^{1/n}$$

Solution: Let L be the value of the limit. Evaluating a limit with an exponent of $1/n$ often requires taking the natural logarithm to convert the problem into a Riemann sum.

$$\ln(L) = \lim_{n \rightarrow \infty} \frac{1}{n} \ln \left(\frac{(2n)!}{n!n^n} \right)$$

First, we simplify the argument of the logarithm:

$$\frac{(2n)!}{n!} = (n+1)(n+2) \cdots (2n)$$

Thus, dividing by n^n gives:

$$\frac{(2n)!}{n!n^n} = \frac{n+1}{n} \cdot \frac{n+2}{n} \cdots \frac{n+n}{n} = \prod_{k=1}^n \left(1 + \frac{k}{n} \right)$$

Taking the natural logarithm of this product converts it into a sum:

$$\ln \left(\prod_{k=1}^n \left(1 + \frac{k}{n} \right) \right) = \sum_{k=1}^n \ln \left(1 + \frac{k}{n} \right)$$

Substitute this back into our limit expression:

$$\ln(L) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \ln \left(1 + \frac{k}{n} \right)$$

This is the definition of the Riemann integral for the function $f(x) = \ln(1+x)$ over the interval $[0, 1]$:

$$\ln(L) = \int_0^1 \ln(1+x) dx$$

We compute this integral using integration by parts, where $u = \ln(1+x)$, $du = \frac{1}{1+x} dx$ and $dv = dx$, $v = 1+x$ (a valid choice of anti-derivative for dx that simplifies the process):

$$\int \ln(1+x) dx = (1+x) \ln(1+x) - \int 1 dx = (1+x) \ln(1+x) - x$$

Evaluating from 0 to 1:

$$[(1+x) \ln(1+x) - x]_0^1 = (2 \ln 2 - 1) - (0 - 0) = 2 \ln 2 - 1 = \ln(4) - 1$$

Equating this back to $\ln(L)$, we find L :

$$\ln(L) = \ln(4) - 1 \implies L = e^{\ln(4)-1} = \frac{4}{e}$$

Since $e \approx 2.718$, we approximate the value:

$$L = \frac{4}{2.718} \approx 1.471$$

The question asks for the integer nearest to this value. Since $1.471 < 1.5$, the nearest integer is:

1

Saturday 20 June 2026

5.1 Daily Limit

Infinite Series Limit of a Sequence

Evaluate the limit:

$$\lim_{x \rightarrow \infty} \sum_{n=0}^x (-1)^n \frac{n+1}{(2n+1)!}$$

Solution: Taking the limit as $x \rightarrow \infty$ of the partial sums exactly forms an infinite series. Thus, the expression is equivalent to:

$$S = \sum_{n=0}^{\infty} (-1)^n \frac{n+1}{(2n+1)!}$$

To evaluate this series, we want to manipulate the numerator $n+1$ to closely match terms in the denominator's factorial. We can rewrite the numerator as:

$$n+1 = \frac{1}{2}(2n) + 1 = \frac{1}{2}(2n+1) + \frac{1}{2}$$

Substituting this into the series allows us to split the sum into two separate infinite series:

$$S = \sum_{n=0}^{\infty} (-1)^n \frac{\frac{1}{2}(2n+1) + \frac{1}{2}}{(2n+1)!} = \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n \frac{2n+1}{(2n+1)!} + \frac{1}{2} \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)!}$$

For the first series, the term $(2n+1)$ cancels with the last multiplier in the factorial $(2n+1)! = (2n+1)(2n)!$:

$$\frac{1}{2} \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n)!}$$

We recognize these two infinite series as the well-known Maclaurin series expansions evaluated at $x = 1$. Recall the series for cosine and sine:

$$\cos(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \implies \cos(1) = \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n)!}$$

$$\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \implies \sin(1) = \sum_{n=0}^{\infty} (-1)^n \frac{1}{(2n+1)!}$$

Substitute these recognized values back into our split series expression:

$$S = \frac{1}{2} \cos(1) + \frac{1}{2} \sin(1)$$

Thus, the limit cleanly evaluates to:

$$\frac{\cos(1) + \sin(1)}{2}$$